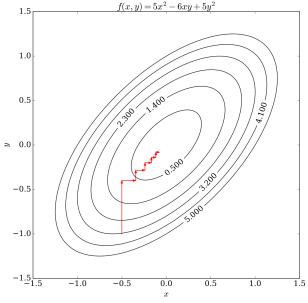
Optimization in Machine Learning

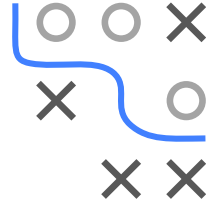
Coordinate descent



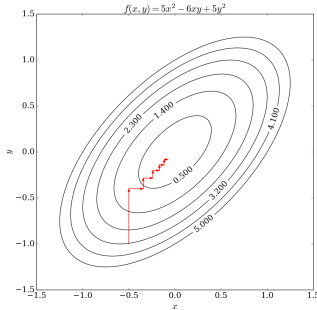
Learning goals

- Axes as descent direction
- CD on linear model and Lasso
- Soft thresholding

COORDINATE DESCENT



- Assumption: objective function not differentiable
- Idea: instead of gradient, use coordinate directions for descent
- First: Select starting point $\boldsymbol{x}^{[0]} = (x_1^{[0]}, \ldots, x_d^{[0]})$
- Step t: Minimize f along x_i for each dimension i for fixed $x_1^{[t]}, \ldots, x_{i-1}^{[t]}$ and $x_{i+1}^{[t-1]}, \ldots, x_d^{[t-1]}$:



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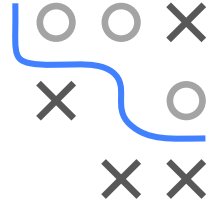
COORDINATE DESCENT



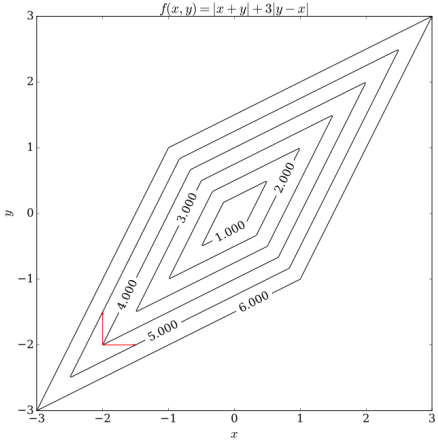
- Minimum is determined with (exact/inexact) line search
- Order of dimensions can be any permutation of $\{1, 2, \ldots, d\}$
- Convergence:
  - f convex differentiable
  - f sum of convex differentiable and *convex separable* function:

$$f(\boldsymbol{x}) = g(\boldsymbol{x}) + \sum_{i=1}^{d} h_i(x_i),$$

  - where g convex differentiable and h_i convex

COORDINATE DESCENT

No convergence in general for convex functions, counterexample:



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EXAMPLE: LINEAR REGRESSION

Minimize LM with L2 loss via CD

$$\min g(\boldsymbol{\theta}) = \min_{\boldsymbol{\theta}} \frac{1}{2} \sum_{i=1}^{n} \left(y^{(i)} - \boldsymbol{\theta}^\top \boldsymbol{x}^{(i)} \right)^2 = \min_{\boldsymbol{\theta}} \frac{1}{2} \|\boldsymbol{y} - \mathbf{X}\boldsymbol{\theta}\|^2$$

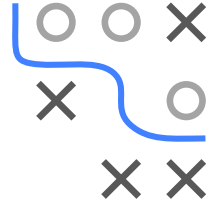
where $\boldsymbol{y} \in \mathbb{R}^n$, $\mathbf{X} \in \mathbb{R}^{n \times d}$ with columns $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_d \in \mathbb{R}^n$

Assume orthogonal design for intuition, i.e., $\mathbf{X}^\top \mathbf{X} = I_d$:

$$\begin{aligned} g(\boldsymbol{\theta}) &= \frac{1}{2} \boldsymbol{y}^\top \boldsymbol{y} + \frac{1}{2} \boldsymbol{\theta}^\top \boldsymbol{\theta} - \boldsymbol{y}^\top \mathbf{X} \boldsymbol{\theta} \\ &\overset{(*)}{=} \frac{1}{2} \boldsymbol{y}^\top \boldsymbol{y} + \frac{1}{2} \boldsymbol{\theta}^\top \boldsymbol{\theta} - \boldsymbol{y}^\top \sum_{k=1}^{d} \boldsymbol{x}_k \theta_k \end{aligned}$$

$^{(*)}$ $\mathbf{X}\boldsymbol{\theta} = \boldsymbol{x}_1 \theta_1 + \boldsymbol{x}_2 \theta_2 + \cdots + \boldsymbol{x}_d \theta_d = \sum_{k=1}^{d} \boldsymbol{x}_k \theta_k$

EXAMPLE: LINEAR REGRESSION



- Exact CD update in direction j:

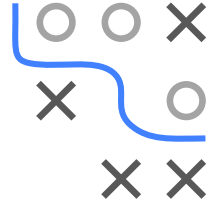
$$\frac{\partial g(\boldsymbol{\theta})}{\partial \theta_j} = \theta_j - \boldsymbol{y}^\top \boldsymbol{x}_j$$

- By solving $\frac{\partial g(\boldsymbol{\theta})}{\partial \theta_j} = 0$ we get

$$\theta_j^* = \boldsymbol{y}^\top \boldsymbol{x}_j$$

- Repeat this update for all θ_j

SOFT THRESHOLDING



Minimize LM with L2 loss and L1 regularization via CD

$$\min_{\boldsymbol{\theta}} h(\boldsymbol{\theta}) = \min_{\boldsymbol{\theta}} \frac{1}{2}\|\boldsymbol{y} - \mathbf{X}\boldsymbol{\theta}\|^2 + \lambda\|\boldsymbol{\theta}\|_1$$

Note that $h(\boldsymbol{\theta}) = \frac{1}{2}\boldsymbol{y}^\top\boldsymbol{y} + \frac{1}{2}\boldsymbol{\theta}^\top\boldsymbol{\theta} - \sum_{k=1}^{d}(\boldsymbol{y}^\top\boldsymbol{x}_k\theta_k + \lambda|\theta_k|)$
Again assume $\mathbf{X}^\top\mathbf{X} = I_d$. Since $|\cdot|$ not differentiable, distinguish cases:

- **Case 1:** $\theta_j > 0$. Then $|\theta_j| = \theta_j$ and

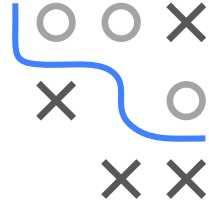
$$0 = \frac{\partial h(\boldsymbol{\theta})}{\partial \theta_j} = \theta_j - \boldsymbol{y}^\top\boldsymbol{x}_j + \lambda \qquad \Leftrightarrow \qquad \theta^*_{j,\text{Lasso}} = \theta^*_j - \lambda$$

- **Case 2:** $\theta_j < 0$. Then $|\theta_j| = -\theta_j$ and

$$0 = \frac{\partial h(\boldsymbol{\theta})}{\partial \theta_j} = \theta_j - \boldsymbol{y}^\top\boldsymbol{x}_j - \lambda \qquad \Leftrightarrow \qquad \theta^*_{j,\text{Lasso}} = \theta^*_j + \lambda$$

- **Case 3:** $\theta_j = 0$

SOFT THRESHOLDING



We can write the solution as:

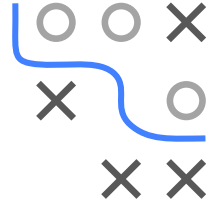
$$\theta^*_{j,\text{Lasso}} = \begin{cases} \theta^*_j - \lambda & \text{if } \theta^*_j > \lambda \\ \theta^*_j + \lambda & \text{if } \theta^*_j < -\lambda \\ 0 & \text{if } \theta^*_j \in [-\lambda, \lambda], \end{cases}$$

This operation is called soft thresholding

Coefs for which solution to unregularized problem is smaller than threshold ($|\boldsymbol{\theta}^*_j| < \lambda$) are shrunk to zero NB: Derivation of soft

thresholding operator not trivial (subgradients)

CD FOR STATISTICS AND ML



Why is it being used?

- Easy to implement
- Scalable: no storage/operations on large objects, just current point
  $\Rightarrow$ Good implementation can achieve state-of-the-art performance
- Applicable for non-differentiable (but convex separable) objectives

Examples:

- Lasso regression, Lasso GLM, graphical Lasso
- Support Vector Machines
- Regression with non-convex penalties